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MANAGING PRODUCT RETURNS FOR REMANUFACTURING*

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Firms are often encouraged to offer environmentally friendly products as a demonstration of corporate citizenship. However, this may prove to be an unrealistic expectation since a rational firm will only engage in profitable ventures; those that increase shareholder wealth. We develop a framework for analyzing the profitability of reuse activities and show how the management of product returns influences operational requirements. We show that the acquisition of used products may be used as the control lever for the management and profitability of reuse activities. These activities, termed product acquisition management, affect several important business decisions. First, if a firm is to pursue reuse activities, these reuse activities must be value-creating. Second, if a firm is to compete by offering remanufactured products, then we show how product returns management influences the overall profitability of such activities via a trial and error EVA® approach. Third, we show how operational issues are strongly affected by the approach used to manage product returns. There is a need for future research specifying the mathematical relationship between acquisition price and the nominal quality of the returned product.
(REMANUFACTURING; ECONOMIC VALUE-ANALYSIS; PRODUCT ACQUISITION MANAGEMENT)

1. Introduction

Recoverable product environments are environmentally conscious systems where products are returned from end users to be reused. Reuse options include value-added recovery (i.e., repair and remanufacture) and material recovery (i.e., recycling) (Thierry, Salomon, Van Nunen, and Van Wassenhove 1995). These options prevent waste by diverting materials from landfills and conserve natural resources (energy and materials). Recoverable product systems are also profitable. In the U.S. there are over 70,000 remanufacturing firms with total sales of $53 billion (USD) (Lund 1998). These firms directly employ 350,000 workers, and average profit margins exceed 20% (Nasr et al. 1998).

In the European Union (EU) a number of recent legislative acts, known as producer responsibility laws, require manufacturers to collect and reuse their products. The requirements for firms doing business in the EU are clear, and these regulations may also act as entry barriers for firms not aware of the changes required for reverse logistics activities. Firms designing products for reuse may have a competitive advantage in lower costs and enhanced

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market share due to the growing consumer awareness and preferences for environmentally conscious products (Kodak 1999; Xerox 1999).

Several authors have cited a need for careful economic analysis of the potential benefits of environmental activities (Esty and Porter 1998; Reinhardt 1999). Firms are often encouraged to offer environmentally friendly products (i.e., recoverable products) as a demonstration of corporate citizenship. However, this may prove to be an unrealistic expectation since a rational firm will only engage in profitable ventures, those that increase shareholder wealth. Further, it may not be reasonable for every original equipment manufacturer (OEM) to engage in reuse activities. The fastest growing firms in electronics and telecommunications may need all the available capital to invest in core activities. In addition, the stock market is expecting high returns from these sectors, and firms may require high return on capital expenditures or favorable economic value analyses (EVA®; EVA® is a registered trademark of Stern Stewart & Co.). It may be rational for an OEM to not engage in reuse activities, to subcontract the reuse operations, or encourage start-up corporate spin-offs to assume the responsibility. The decision whether or not to engage in reuse activities directly, indirectly, or not at all should be driven by a thorough economic analysis of the costs and benefits of such a program. In the event that recovery is mandated, a careful economic analysis is required to determine the best way to do the recovery.

Despite the potential profitability of reuse and the legislative requirements, there are no integrated decision frameworks to advise decision-makers about the economic viability of various reuse options. We seek, in part, to develop the foundation for analyzing the potential profitability of reuse options. Specifically we develop a framework for analyzing the profitability of reuse activities and show how the management of product returns influences operational requirements. The framework is based on our experiences with companies that presently use an ad hoc trial and error approach for analyzing the potential of reuse activities. We intend the framework to serve as a structured method for examining reuse alternatives and to serve as the foundation for further research that would lead to explicit relationships and perhaps closed-form optimizations within our framework.

A firm operating a recoverable product system relies on the return of used products as the basic input (raw material) to a reuse system (repair, remanufacturing, cannibalization, recycling). The return rate of used products has been regarded as an exogenous process that cannot be controlled by the firm. We hypothesize that return rates may be influenced by actions taken by the firm, and this decision, when appropriate, should be a conscious choice. Product acquisition management addresses several essential issues (see Figure 1). First, we intend to show that product acquisition management determines whether reuse is a value-creation activity for a specific firm. Second, if reuse activities are profitable, to maximize revenue the appropriate method for managing product returns should be selected. Third, operational issues, such as facility design, product planning and control policies, and inventory policies, are dependant on the method selected to manage product returns. Fourth, product acquisition management activities help show the potential economic benefits of identifying and developing new markets for reused product, and to balance the return rates with market demands. We will focus on the first three issues in this research and recognize that while the development of new markets is crucial to the success of product reuse, it is not the focus of our present research.

This view represents a fundamentally new approach to the economics and operations of reuse activities. Traditionally, research efforts have focused on ways to minimize the loss associated with reuse; e.g., cost minimization models. This traditional view of reuse activities requires that firms seek to create new markets for used products, but passively accept product returns. We propose a structured method for valuing the potential economic attractiveness of reuse activities.

In this manuscript we focus on the concept of product acquisition management, and how this concept may be used in several contexts. Product acquisition management is a key input to assessing the potential economic attractiveness of reuse activities, and as a foundation for operational planning and control activities. We reject the idea that firms must passively

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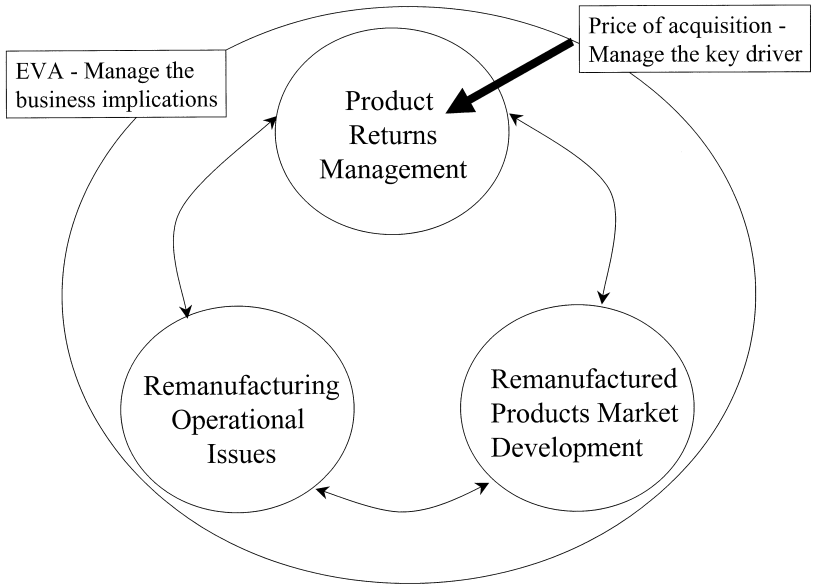


FIGURE 1. Product Acquisition Management.

accept product returns, and show that a system for control exists. The idea that a firm may control the quality, quantity, and timing of product returns has a profound impact on the understanding of how reuse systems may be profitable in some cases. We also show that this new approach leads to many research issues that remain unanswered.

We begin with a literature review in the following section. In subsequent sections, we present an overview of the product returns process and choices, a detailed discussion of product acquisition management activities, and a discussion of research needs.

2. Research Literature

We refer the reader to Fleischmann et al. (1997), and Guide, Jayaraman, Srivastava, and Benton (2000) for complete literature reviews. We note that much of the previous research concerned with reuse activity focuses on a particular functional area or activity, such as network design (Spengler, Püchert, Penkuln, and Rentz 1997; Krikke 1998; Krikke, Van Harten, and Schuur 1999), shop floor control (Guide, Srivastava, and Kraus 1997; Guide and Srivastava 1998), or inventory control (Inderfurth 1997; van der Laan 1997; van der Laan, Salomon, Dekker, and Van Wassenhove 1999). No published research explicitly considers the problem of managing or controlling products returns. The research literature has a number of restrictive assumptions. First, all of the models assume that product returns are an exogenous process. An exogenous process, in this case, refers to the assumption that return rates, quality, and timing are all outside the direct control of the firm. The firm passively accepts all returns, and the returns process itself is a black box operation. Second, when product returns are explicitly considered, return rates are assumed independent of sales rates. The sole exception to this assumption is Kelle and Silver (1989), who assume a time-lagged return rate. Finally, although several authors (Fleischmann et al. 1997; Guide, Jayaraman, Srivastava, and Benton 2000) have called for a more integrated approach to logistics planning, no such research has appeared.

Under the classic assumption of exogenous, uncontrollable return rates and quality, there is research showing that remanufacturing operations are more complex to plan, manage, and control than traditional manufacturing operations (Guide, Srivastava, and Kraus 1997; Guide and Srivastava 1998). The primary reason for this complexity is the high degree of variability in the quality of used products that serve as raw materials for the production process. This

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variability makes the tasks of materials planning, capacity planning, scheduling, and inventory management complex and difficult to manage. Managing this high amount of variability is expensive for the firm since decoupling the system requires higher investments in materials, equipment, and labor.

We also note there is evidence that some remanufacturing firms actively control the quality, quantity, and timing of product returns (Guide 2000). This evidence shows that the classic assumption that a firm must passively accept product returns with no ability to influence returns quality, quantity, and timing is not valid. Given the high cost of complexity, it is logical that firms have developed methods for controlling the product returns process. In the following section we discuss product returns and provide detailed examples of product acquisition management.

3. Product Returns

In the past, the EU has been a diverse set of differentiated markets driven by many different cultures and tastes. This market segmentation has made product reuse difficult because of the lack of volume of standardized consumer goods. However, with the adoption of the Euro as a common currency, unified markets are rapidly developing across the EU in some sectors that are not cultural, e.g., consumer electronics (PCS, cellular telephones, personal stereos), and automobiles. The consumer electronics markets have the greatest growth potential for reuse activities since volumes are high and product life cycles are increasingly short. In general, several trends have made producer responsibility legislation increasingly popular. Millions of products are produced annually with shorter product life cycles, and this leads to a huge number of obsolete products entering the waste stream. The final disposition of these products poses an enormous problem for the environment, and a large potential stream of used products for reuse operations.

Legislation in the United States tends to encourage, rather than mandate, reuse activities. The recycling industry in the United States is an example of such a system where legislation encourages reuse via tax credits, or municipalities assume collection responsibilities. However, individual states have banned the landfill of cathode ray tubes and some electronics equipment, and the number of states banning specific types of products from landfill is expected to grow.

There are two primary systems for obtaining used products from the end-users for reuse: the waste stream system and the market-driven system. The waste stream system relies on diverting discarded products from landfills by making producers responsible for the collection and reuse of their products. A market-driven system relies on financial incentives to motivate end-users to return their products to a firm specializing in the reuse of those products.

In the waste stream system, firms passively accept all product returns from the waste stream. Unable to control the quality of returns, firms often consider the large volumes of returns a nuisance, and naturally tend to focus on the development of low cost reverse logistics networks. The result of the product returns mandates and policies is a large uncontrolled volume of used products flowing back in increasing volumes to the original equipment manufacturers. Firms are ill prepared to cope with the complexity of product returns and end-of-life disposition and are seeking ways to minimize their losses.

In the European Union (EU) a number of recent legislative acts, known as producer responsibility laws, require manufacturers to assume responsibility for the end-of-life disposition of their products (Guide, Jayaraman, Srivastava, and Benton 2000). For instance, in Germany, the Commercial and Industrial Waste Avoidance and Management Act (KrW/AbfG) requires that the producer is responsible for the end-of-life disposition of their products, including reuse and recovery. The Ministerial Bill of Ordinance on Disposal of Information Devices (IT-Geräte-V) is an agreement whereby manufacturers of electrical and

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electronic equipment (specifically information technology devices) take responsibility for the recovery and end-of-life disposition of their products (Gotzel, Weidling, Heisig, and Inderfurth 1999). In the Netherlands, the electronics waste act has similar requirements. Some legislative acts may require the end-user to return the used product to a collection system for reuse. The used car ordinance in Germany requires that the last owner of an automobile deliver it to a collection center (Gotzel, Weidling, Heisig, and Inderfurth 1999). The requirements for firms doing business in the EU are clear, and these regulations may act as entry barriers for firms not aware of the changes required for reverse logistics activities.

In a market-driven system, end-users are motivated to return end-of-life products by financial incentives, such as deposit systems, credit toward a new unit, or cash paid for a specified level of quality. Firms are able to control the level of quality of returned products since acceptance of returns is conditioned by standards. Market-driven systems are common in the United States because of the profitability of remanufacturing (Guide 2000). Firms using a market-driven approach for product returns focus mainly on high value industrial products (Nasr, Hughson, Varel, and Bauer 1998; Guide 2000). However, there are exceptions; the system for remanufacturing automotive parts has been well established since the 1920s. As the auto parts industry illustrates, there is also money to be made with high volume, low value products [see Kodak's (1999) description of its single-use camera program for another example of high volume, low value products]. A growing number of firms are interested in the consumer electronics segment, and this is the largest growth area in remanufacturing in the United States.

A combination of the market-driven and waste stream approaches is also possible. Product returns may be mandated or encouraged by legislative acts, but firms may still encourage the returns of products in known condition by offering incentives. A firm using a pure waste stream approach or a pure market-driven approach will have facilities with different operational characteristics and managerial control problems.

In contrast, the two systems result in extremely different views of reuse activities. In the waste stream system cost reduction is encouraged, and the fundamental issue is to minimize the amount of money the firm loses. The market-driven system views reuse as a profitable economic proposition. Past research has focused almost exclusively on the waste stream approach to reuse activities, and it is logical that the modeling efforts are aimed at cost minimization. In the following section, we examine the operational implications of the two pure product acquisition management approaches.

4. Impact of Product Returns on Operational Characteristics

The waste stream and market-driven approaches lead to facilities with different operational characteristics. In the waste stream approach, Case 1 (Figure 2), the first significant difference is the lower quality, on average, of used products arriving to the facility. These used products also have large variances in quality, quantity, and timing. Since there is no effort made to control the incoming quality of used products, additional facility space is required to store, sort, and grade the returned products. A large number of units will need to be disposed of since a high proportion of units will not be recoverable for remanufacturing. Incoming products to the remanufacturing shops will have high variability in routings and in required processing times. This high variability will lead to long queues at work centers, unpredictable lead times, and low throughput. The overall high levels of variability make this remanufacturing facility difficult to plan, control, and manage (Guide, Srivastava, and Kraus 1997; Guide and Srivastava 1998). Operating expenses will be greater since additional labor and machine costs are required. The facility will most likely resemble the process based, classic repair job shop with jumbled product flows. These types of shops are difficult to plan, manage, and control.

A firm actively controlling the incoming level of quality, Case 2, by using a market-driven

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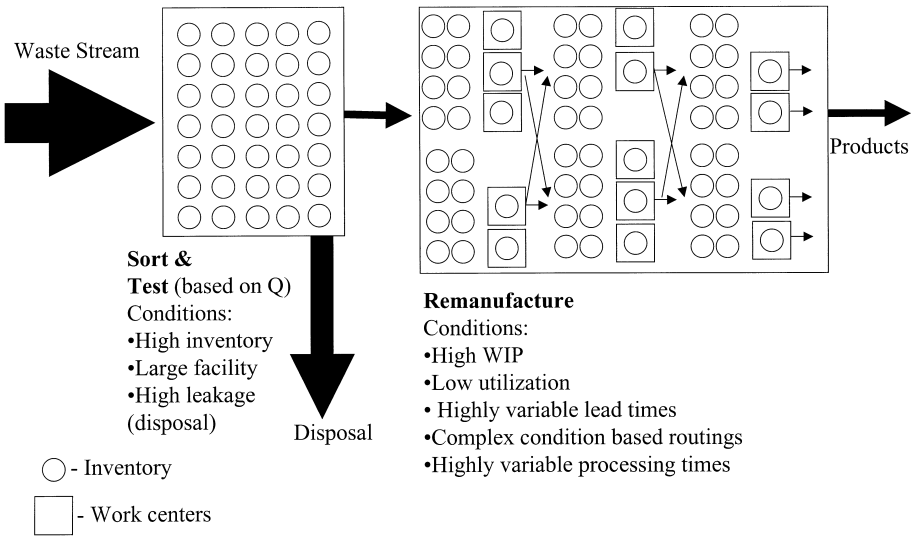


FIGURE 2. Case 1: Waste Stream.

approach (see Figure 3) has several significant operational benefits. Products will be sorted and graded before entering the remanufacturer's possession. The size of the facility reserved for arriving used products will be less than the waste stream case since products will be tested to ensure that quality standards are met. Lower amounts of used product inventory will be needed, and disposal costs will be decreased. The variability of routings and processing times required will be reduced because of the higher levels of incoming quality. This lower variability will result in shorter queues, better machine and labor utilization, and more predictable flow times. Throughput for the facility will increase with no additional machines or labor, and machine and/or labor costs may be lower for higher volumes of output. The facility layout appears as a flowshop, with dominant product flows made possible by more standardized routings. When a market-driven approach is used, the overall task of planning,

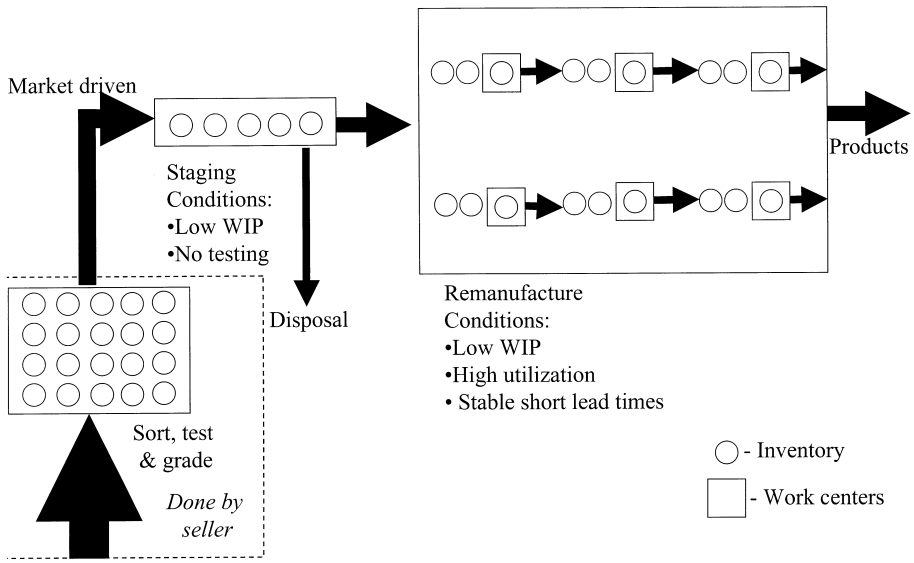


FIGURE 3. Case 2: Market Driven.

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controlling, and managing a facility is simplified. This should lead to a more productive and cost-effective system.

We suggest that market forces will drive companies to evolve management systems that allow greater control. This natural evolution is shown in the example that follows, referring to a cellular telephone remanufacturer, that has developed a method for obtaining used products that allows greater control over the profitability of their operations.

5. Product Returns at ReCellular, Inc.

ReCellular, Inc., is a third-party remanufacturer of mobile telephones located in the United States. The company has been in business for almost 10 years and has remanufactured over one million phones since its inception. The process of product acquisition management has evolved from a system of buying discarded mobile phones in bulk from cellular airtime providers, to a system of offering established prices for phones with a specific nominal level of quality. The information in this section was gathered through in-depth interviews with the president of ReCellular, Charles Newman.

The original system of acquiring used mobile phones involved getting cellular airtime providers to collect and send the discarded phones to ReCellular. The phones were acquired for a minimal acquisition price, but there were other substantial costs. This method required an extensive sorting, testing, and storage area since the phones arrived in unknown condition. Labor costs were high because of the labor-intensive tasks of sorting and grading. Phones often were sold as scrap or disposed of since the quality level was poor and the cost to remanufacture high. Many phones were missing recharging systems and had physical damage. Gradually the system has evolved to the present method of setting a price for a known level of quality (see Tables 1 and 2). Each seller that ReCellular purchases cellular telephone from has an established level of nominal quality, and prices are based, in part, on this level of quality. The seller's level of quality is relatively consistent based on the source of used cellular telephones. ReCellular's purchasing agents must spend a substantial amount

TABLE 1

Nominal Quality Metrics for Grading Mobile Telephones (*ReCellular, Inc.*)

Nominal Quality Level	Description
1	Passes a functional and electrical test and with minimal cleaning, produces a nice looking phone. Phones show no water or physical damage and have a good antenna, display, lens, keypad, and a working primary charging system.
2	Passes a functional and electrical test and with cleaning and buffing can produce a phone that is acceptable in appearance. Phones show no water or physical damage and have a good antenna, display, lens, keypad, and a working primary charging system. Keypad and plastic may show some fading or uniform discoloration.
3	Passes a functional and electrical test but will require some amount of rework to make this phone useable. Phone does have a good display and working keypad. Phones may have some physical damage and may show signs of wear that will require the replacement of some parts.
4	Phones may have the same physical appearance as a grade 1 phone, but the lens may have some scratches. These phones failed the test or did not power up. Many of these phones are repairable and should produce a good yield with some technical knowledge.
5	Phones have the same physical appearance as a grade 2 phone, but failed the test or did not power up. Many of these phone can be repaired and should produce a good yield with some technical knowledge.
6	Phones have failed the test or did not power up. Phones have some damage but can be used for part salvage.

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TABLE 2
Prices Offered for Specific Mobile Phones with Specific Levels of Nominal Quality

Preliminary Pricing	1	2	3	4	5	6
Ericsson DH336/338	$ 33	$ 31	$ 29	$25	$20	$15
Motorola Startac 3000	$ 85	$ 75	$ 65	$55	$35	$25
Nokia 5100/6100 series	$110	$105	$100	$90	$80	$60
Sony CMRX100	$ 40	$ 35	$ 30	$25	$15	$10

of time with a seller in the beginning to explain how phones are graded and priced. The seller is then responsible for ensuring that the phones are of acceptable levels of quality before sending them to ReCellular.

The shift to a more focused, intelligent purchasing system has lead to a number of benefits for ReCellular. This product acquisition management strategy has enabled the firm to reduce the need to sort and grade phones in-house. It has also served to reduce the level of variability for inputs to the system, making many management tasks more predictable. In short, ReCellular has moved from a Figure 2 situation to a Figure 3 situation.

An analogy may be drawn with the evolution of quality management practices. Prior to the advent of total quality management systems and just-in-time systems, firms attempted to screen the level of incoming quality of parts and materials by the use of sampling and inspection schemes, and setting acceptable standards for vendor's average outgoing quality levels. When ReCellular began remanufacturing cellular telephones, there was no attempt to control incoming levels of quality. The lots were screened at ReCellular, and cellular telephones of unacceptable quality were disposed of or sold to scrap dealers. Since the Japanese quality movement, firms focus more on preventing quality problems rather than inspection. Firms have gone from systems with percentage defective to parts-per-million defective, with no inspection. This has allowed the direct, synchronized, feed of parts to production lines with no intermediate storage. The quality movement has made systems with shorter lead times and less work-in-process possible. The shift in product acquisition management practices at ReCellular may be seen as evidence of rapidly evolving management control systems in remanufacturing and reuse.

6. Management of Product Returns

As can be seen from the cellular telephone example, it is possible for a firm to manage the quality of product returns by offering financial incentives. In our example, the remanufacturer may choose not to buy cellular telephones of a lower quality. The seller may respond by offering the lower quality cellular telephones for a lower price, or finding a buyer that will accept the lower levels of quality. This makes the development of cascade reuse systems possible. A seller (broker) of used products may grade the returned products and price the product accordingly. Firms that require used products as inputs for reuse operations then select the price and quality that enables the firm to make an acceptable level of profit. Firms with lower financial rates of return will be able to use products that may be unacceptable (unprofitable) for firms requiring higher rates of return.

There are many types of incentive systems used by firms in the U.S. remanufacturing sector, including deposits, credit toward a remanufactured or new unit, cash for product returns, and leasing (Guide 2000). The product returns management process allows a planner to set a price the firm is willing to pay for a used product. This acquisition price is the direct cost to the firm and will be greater than zero, but presumably less than the cost of a new product. There are other indirect costs, such as the logistics costs, which include collection, transportation, storage, and sorting. A firm may offer an attractive enough incentive so that the end-user is

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motivated to return the product to a reuse site, thus eliminating collection costs. The cost of logistics in this case will be greatly reduced. Product sorting will occur at the time a used product is offered, and firms will not accept products below a minimal quality level. A market-driven approach will encourage the formation of cascade reuse systems, where products unprofitable for one firm to remanufacture may be offered to firms willing to recycle materials.

In order for market-driven returns to operate effectively, firms will need to develop tests to measure product quality; otherwise, consumers will be motivated to return inferior quality products for a premium payback; the higher the value of the used item, the greater the incentive for a firm to develop a better inspection system. Firms and consumers may benefit from the development of formal bidding systems that match reuse firms' needs with potential sources of used products. The Internet is a logical location for an on-line bidding system focused on product reuse, and several such sites are presently in operation (e.g., http://www.buyreman.com).

Products obtained from the waste stream will have no direct cost for obtaining the products; however, there are logistics costs associated. Countries mandating producer responsibility ensure that firms will have sufficient quantities of product returns, but the producer bears the cost associated with developing and maintaining a reverse logistics network. The quality of product that is returned via the waste stream is highly uncertain since individual users must decide when to return a product for reuse. Presumably, products returned as part of the waste stream will be of lower average quality than products returned from a market-driven system. The problem becomes one of sorting products that may be economically reused from products that are only fit for recycling or landfill. Secondary sorting may be required when only parts from selected end items may be reusable. Firms must maintain large amounts of inventory prior to sorting and provide a large enough facility for storage of the returned items.

We hypothesize a given price yields a distribution of nominal quality of the returned products. By a distribution of nominal quality, we mean that classes of quality are defined, and returns may be graded so as to fit into one of the quality classes, and a frequency distribution fit to the number of items in each of the quality classes. There is evidence that product returns have a distribution of nominal quality (Krikke, Van Harten, and Schuur 1999). The nominal distribution of quality is conditional on the acquisition price paid for the used product, although the exact relationship is unspecified at this time. The cost to remanufacture a used product is inversely related to the cost of acquisition since the quality of the used product will determine the amount of material replacement and labor content, among others. By examining various levels of average nominal quality and the associated acquisition cost, a firm may be able to determine what minimum level of quality is necessary for reuse to be economically attractive.

The selling price for a remanufactured good may be constrained by the market's willingness to purchase remanufactured goods. A firm may determine the profit margin for a given level of nominal quality. This information may be used for an economic value analysis to determine whether remanufacturing is economically feasible.

7. Economic Value Added

Reuse activities may not be economically attractive for all firms, and a manager should be able to determine the most attractive option for their firm. Economic Value Added (EVA) provides a robust method for determining the potential profitability of reuse opportunities. EVA measures the difference between the return on a company's capital and the cost of that capital (Young 1997). A positive EVA indicates that value will be created for the firm's shareholders that satisfies their expectation; a negative EVA shows that value will not be created.

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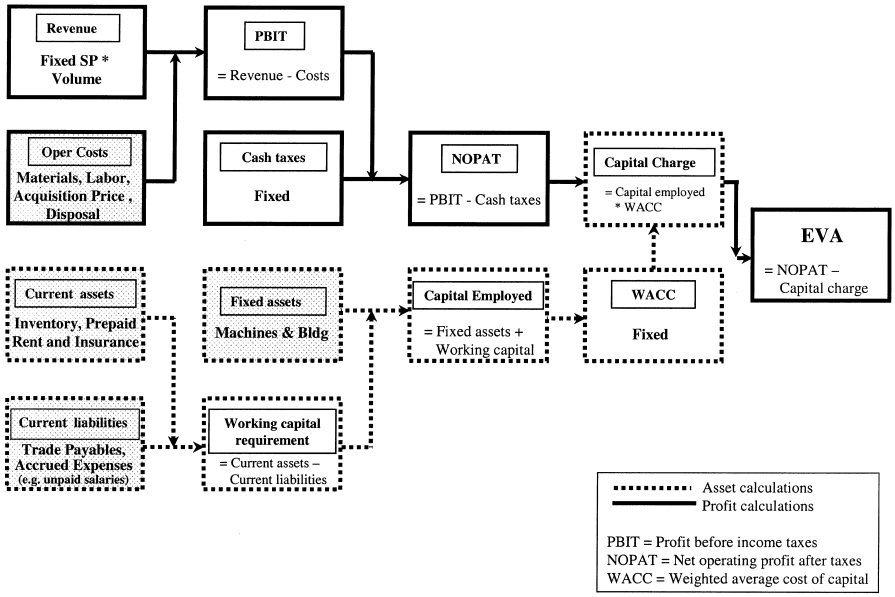


FIGURE 4. EVA Calculations.

We posit that EVA can be of benefit to a manager in examining whether sufficient value is created via reuse activities, and that EVA may be used to determine the best product acquisition management decisions. We assume that the market for remanufactured products exists and that we know the price/quantity relationship, i.e., for a given price a remanufacturer knows how many remanufactured products will be sold, and commits to deliver this quantity. We can now use an EVA what-if analysis for this situation where revenue is fixed. We will show below that the acquisition price is a driving variable in the level of the EVA.

Figure 4 shows the flow of decisions in our EVA calculations. The shaded boxes are affected by alternative product acquisition management decisions. (Note that for reasons of clarity we have chosen to keep our EVA discussion relatively crude and simple). The key to controlling the potential profitability is the quality of used products the firm desires (see Figure 5). By

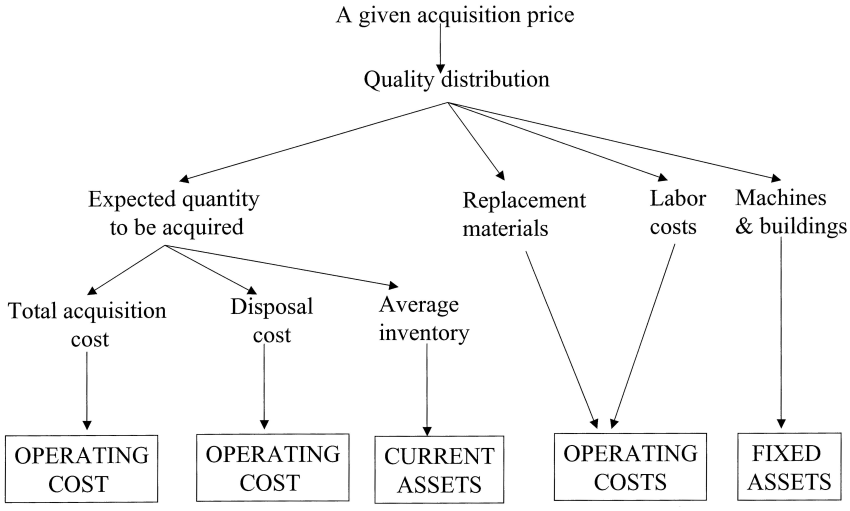


FIGURE 5. A Simple View of the Impacts of Acquisition Price.

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changing the minimal acceptable quality level for used products, the price for acquisition changes. The price of acquisition acts as a driver for many of the cost relationships. The price of acquisition directly affects operating costs (via the costs of acquisition, labor, replacement materials, and disposal costs). A higher nominal quality of returned products will decrease the labor content, the replacement materials, and the disposal of products that are not economically recoverable. However, increasing the price paid for products may increase operating expenses as higher prices are paid for higher levels of quality. A higher level of nominal quality may also decrease a firm's investment in fixed assets (via machines and buildings). Since the condition of products is more predictable with a higher level of quality, less storage space and machine capacity are required (see Figures 2 and 3). The impact of quality on current assets is less clear since higher quality will affect inventories in two ways. First, the total amount of inventory held by the system will decrease due to lower variance. Second, the value of acquired products (the raw materials) will increase because of higher acquisition costs. Of course, current assets and liabilities may also be influenced by decisions like buying or renting (e.g., storage space).

In any case, it is not conceptually difficult to analyze the relative EVA impact of product acquisition management alternatives, given the approximate relationship between nominal quality and acquisition price. The EVA shows, with minimal information, whether or not a firm should consider remanufacturing activities. The impact of using a wide range of levels of nominal quality may be simulated since each level of quality has an associated acquisition price. The challenge is to determine the exact relationship between the cost of acquisition and the levels of nominal quality. At present, the exact relationships are unknown, and managers are currently using simple ad hoc estimates. However, these estimates provide managers with enough information to run rough EVA calculations. The Hewlett-Packard Company (Davey 1999) presently uses a trial and error EVA approach to analyze the potential profitability of reuse activities. The manager uses pessimistic, neutral, and optimistic estimates in doing the calculations and simulates the regions around each of these estimates to get an answer that is reasonably robust to changes in the input values. Companies may not be able to exactly specify the functional relationships we propose using, but the managers are capable of defining a number of reasonable alternatives and making estimates based on these. Experienced managers possess expert knowledge that makes these approximations acceptable in a business setting. A manager can simulate the impact of various scenarios to determine whether remanufacturing is an attractive option for a firm, and determine what investment in quality is optimal for the individual organization.

A firm may lower the costs of reuse and choose to lower the selling price. By changing the selling price, we change the quantity sold, and hence the overall EVA may be used to examine a large number of scenarios by altering the input (acquisition) as well as the output (market) side. If enough information is known about the relations shown in Figure 5 to specify the mathematical relationships, then an overall optimization may be done for the decision process. The specifying of the mathematical relationships is a topic for further empirical and modeling research that would lead to explicit relationships and perhaps closed-form optimizations within the EVA-product acquisition management framework we propose.

Since individual firms have different requirements for the weighted average cost of capital (see DeBono 1997 for a complete discussion), the decision may be unprofitable for a large firm in a market where high returns are required by stockholders, but profitable for smaller firms where lower returns may be acceptable. The EVA approach explains why for some firms reuse is very profitable, and for others is not economically attractive. EVA also shows that the price of acquisition may be a powerful control lever on the potential profitability of the reuse systems.

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8. Implications and Future Research

There are strategic implications of product reuse activities that are not addressed by the EVA, and they deserve consideration. A firm must consider the risks inherent in not controlling the reuse of their products. Disassembly of used products provides essential information about the durability and reliability of products during the life of the product. This information may be used in developing the next generation of higher quality products, and to assist designers in dematerialization efforts. The confidentiality of product characteristics may produce information asymmetries and competitive advantages. The authors know of one remanufacturer that provides these product details back to the OEM in a cooperative agreement.

A firm must also consider the competitive advantages in adopting a green image. At present, very little research showing the effects of a green image on sales is available. An additional marketing-related strategic concern is the cannibalization of new sales. No studies have been done showing the effect of remanufactured sales on new product sales. There are risks in providing remanufactured products that may compete for sales, and these risks should be better understood.

We recognize that our approach to EVA could be refined. The functional relationships detailed in the previous section should be specified mathematically. This would allow a careful optimization of the EVA process. Other refinements are possible and encouraged. Operational implications require a better understanding of the relationships between the costs of remanufacturing, quality, and acquisition price. Specifically we need to be able to specify the functions for acquisition prices, quality, and remanufacturing costs so that optimization models may be developed.

The areas shown in Figure 1 all deserve close attention. Market development is an unexplored area and little is known about consumer preferences and attitudes toward remanufactured products. The development of secondary markets for remanufactured products (e.g., selling remanufactured laser printers to universities and school districts) also deserves close attention. A strategic question is whether remanufacturing provides product differentiation and the competitive advantages that accompany it. Additionally, firms must consider the interaction between remanufactured product sales and the product life cycle of the product itself.

Managing product returns requires a better understanding of price as an incentive to return products. Present research on quality and price is structured from the manufacturer's point, not the end-user. Another consideration is whether a given acquisition price will result in sufficient quantities of returned products.

Models that consider the dynamics between the areas of market development, product returns management, and operational issues should be carefully considered. A hierarchical approach to planning and control in this environment could produce an insightful managerial tool. Other optimization models incorporating search algorithms may also be useful in this type of environment.

Given the lack of detailed information and theory available, the first requirement is for sound empirical research to better define the relationships and cost functions. Only after a thorough understanding of the actual conditions may refined models be developed. We believe that product acquisition management represents an evolution of the understanding of reuse systems. The firms that emerge as industry leaders will need to develop a quantifiable model for the business of reuse, and we believe this work is a first step in this direction.

9. Conclusions

We have presented a fundamentally new approach to the economics and operations of reuse activities. Traditionally, research efforts have focused on ways to minimize the loss associated with reuse; e.g., cost minimization models. This traditional view of reuse activities requires that firms seek to create new markets for used products, but passively accept product

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returns. We proposed a framework, based on the EVA concept, to value the potential economic attractiveness of reuse activities. The concept of returned product quality acts as a driver in our framework.

We also have shown that product acquisition management may serve as a foundation for operational planning and control activities. We reject the idea that firms must passively accept product returns, and show that a system for control exists using returned product quality as the system regulator. The idea that a firm may control the quality, and quantity of product returns has a profound impact on the understanding of how reuse systems may be profitable in some cases. We also show that this new approach leads to many research issues that remain unanswered.

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